

Tecniche Sperimentali per la Fisica delle Alte Energie

Geant4 – Analisi Esempi + Esercitazione

S. Germani – stefano.germani@unipg.it

Sommario

- Analisi Esempio B1
- Analisi Esempio B4
 - Esempio B4a
 - Esempio B4c
- Esercitazione (Sviluppo Applicazione Indipendente)

Esempi Geant4

https://geant4-userdoc.web.cern.ch/Doxygen/examples_doc/html/index.html

Geant4 Examples

This module collects three sets of user examples aimed to demonstrate to the user how to make correct use of the GEANT4 toolkit by implementing in a correct way those user-classes which the user is supposed to customize in order to define his/her own simulation setup.

The "**basic**" set of examples is oriented to novice users and covering the most typical use-cases of a Geant4 application with keeping simplicity and ease of use.

An "**extended**" set of examples may require some additional libraries besides of Geant4. This set covers many specific use cases for actual detector simulation.

An "**advanced**" set of examples covers the use-cases typical of a "toolkit"-oriented kind of development, where real complete applications for different simulation studies are provided; may require additional third party products to be built.

Prova Esecuzione Esempio

Documentazione Esempi:

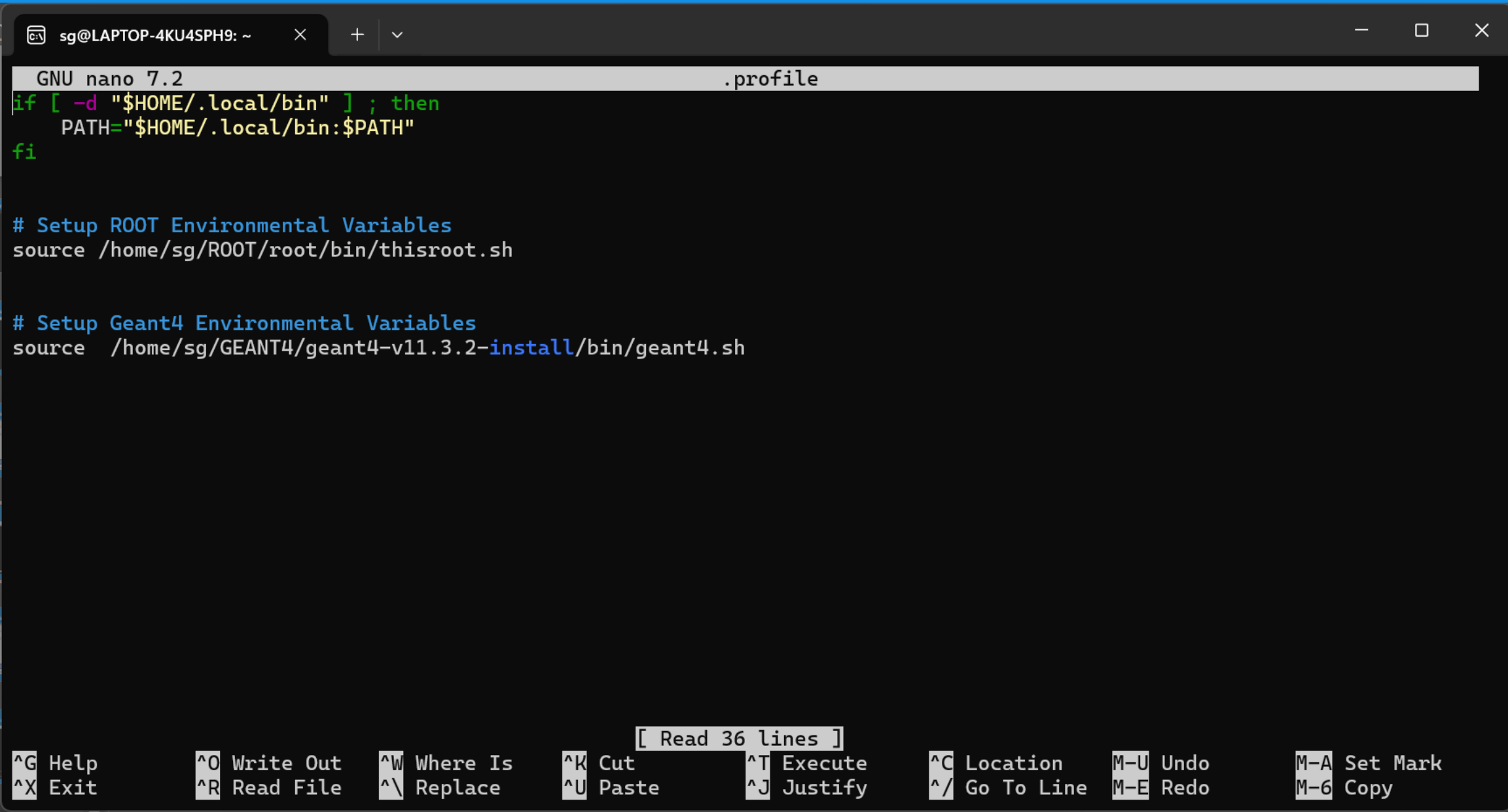
https://geant4-userdoc.web.cern.ch/Doxygen/examples_doc/html/index.html

How to build and run an example

Compile and link to generate an executable

```
% cd path_to_exampleXYZ      # go to directory which contains your example
% mkdir exampleXYZ_build
% cd exampleXYZ_build
% cmake -DGeant4_DIR=path_to_Geant4_installation/lib[64]/cmake/Geant4/ ../exampleXYZ
% make -j N exampleXYZ       # "N" is the number of processes
% make install               # this step is optional
```

Variabili Ambientali



```
GNU nano 7.2 .profile
if [ -d "$HOME/.local/bin" ] ; then
    PATH="$HOME/.local/bin:$PATH"
fi

# Setup ROOT Environmental Variables
source /home/sg/ROOT/root/bin/thisroot.sh

# Setup Geant4 Environmental Variables
source /home/sg/GEANT4/geant4-v11.3.2-install/bin/geant4.sh
```

[Read 36 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy

Esempio B1

Esempio B1

- **Example B1**
 - Simple geometry with a few solids
 - Geometry with simple placements (`G4PVPlacement`)
 - Scoring total dose in a selected volume user action classes
 - Using `G4Accumulable` for automatic merging of scored values in multi-threading mode
 - Geant4 physics list (`QBBC`)

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gt × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$ mkdir B1_build|
```


Prova Esempio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gi × + v
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1$ cd B1_build/
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ pwd
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ls /home/sg/GEANT4/geant4-v11.3.2-install/lib/cmake/Geant4/
G4EXPATShim.cmake      G4MotifShim.cmake      Geant4ConfigVersion.cmake      Modules
G4FreetypeShim.cmake  G4X11Shim.cmake        Geant4LibraryDepends-release.cmake  PTL
G4HDF5Shim.cmake      Geant4Config.cmake      Geant4LibraryDepends.cmake      UseGeant4.cmake
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ cmake -DGeant4_DIR=/home/sg/GEANT4/geant4-v11.3.2
-install/lib/cmake/Geant4/ ../|
```

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ make -j 4 exampleB1  
[ 12%] Building CXX object CMakeFiles/exampleB1.dir/exampleB1.cc.o  
[ 37%] Building CXX object CMakeFiles/exampleB1.dir/src/DetectorConstruction.cc.o  
[ 25%] Building CXX object CMakeFiles/exampleB1.dir/src/ActionInitialization.cc.o  
[ 50%] Building CXX object CMakeFiles/exampleB1.dir/src/EventAction.cc.o  
[ 62%] Building CXX object CMakeFiles/exampleB1.dir/src/PrimaryGeneratorAction.cc.o  
[ 75%] Building CXX object CMakeFiles/exampleB1.dir/src/RunAction.cc.o  
[ 87%] Building CXX object CMakeFiles/exampleB1.dir/src/SteppingAction.cc.o  
[100%] Linking CXX executable exampleB1  
[100%] Built target exampleB1  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ |
```

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gt × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ls  
CMakeCache.txt  Makefile          exampleB1      exampleB1.out  run1.mac    vis.mac  
CMakeFiles      cmake_install.cmake  exampleB1.in  init_vis.mac   run2.mac  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ |
```

Prova Esemplio B1

```
sg@LAPTOP-4KU4SPH9: ~/Gt × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ls  
CMakeCache.txt  Makefile          exampleB1      exampleB1.out  run1.log       run2.mac  
CMakeFiles      cmake_install.cmake exampleB1.in   init_vis.mac   run1.mac       vis.mac  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ more run1.mac |
```


Ispezione Macro Esempio B1

```
sg@LAPTOP-4KU4SPH9: ~/GI  ×  +  ▾  
#  
# Can be run in batch, without graphic  
# or interactively: Idle> /control/execute run1.mac  
#  
# Change the default number of workers (in multi-threading mode)  
#/run/numberOfThreads 4  
#  
# Initialize kernel  
/run/initialize  
#  
/control/verbose 2  
/run/verbose 2  
/event/verbose 0  
/tracking/verbose 1  
#  
# gamma 6 MeV to the direction (0.,0.,1.)  
#  
/gun/particle gamma  
/gun/energy 6 MeV  
#  
/run/beamOn 5  
#  
# proton 210 MeV to the direction (0.,0.,1.)  
#  
/gun/particle proton  
/gun/energy 210 MeV  
/tracking/verbose 2  
#  
/run/beamOn 1  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$
```

Verbosità Output

Definizione Particella Primaria
+ Run di 5 eventi (6 MeV γ)

Nuova Particella Primaria
+ Nuovo Livello Verbosità
+ Nuovo Beam On (1x protone da 210 MeV)

Prova Esemplio B1

[illegible]

```
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ./exampleB1 run1.mac
```

```
*****  
Geant4 version Name: geant4-11-03-patch-02 [MT] (25-April-2025)  
<< in Multi-threaded mode >>  
      Copyright : Geant4 Collaboration  
      References : NIM A 506 (2003), 250-303  
                  : IEEE-TNS 53 (2006), 270-278  
                  : NIM A 835 (2016), 186-225  
      WWW : http://geant4.org/  
*****  
  
<<< Reference Physics List QBBC  
Accumulable registered as "accumulable_0"  
Accumulable registered as "accumulable_1"  
Visualization Manager instantiating with verbosity "warnings (3)"...  
Visualization Manager initialising...  
Registering graphics systems...  
  
You have successfully registered the following graphics systems.  
Registered graphics systems are:  
  ASCIIITree (ATree)  
  DAWNFILE (DAWNFILE)  
  G4HepRepFile (HepRepFile)  
  RayTracer (RayTracer)  
  VRML2FILE (VRML2FILE)  
  gMocrenFile (gMocrenFile)  
  TOOLSSG_OFFSCREEN (TSG_OFFSCREEN, TSG_FILE, TSG)  
You may choose a graphics system (driver) with a parameter of
```

```
G4 kernel has come to Quit state.  
Deleting G4Run (id:1)  
UserDetectorConstruction deleted 0x5ee3f71aff80  
UserPhysicsList deleted 0x5ee3f71b0c10  
UserActionInitialization deleted 0x5ee3f736c000  
UserWorkerInitialization deleted 0  
UserWorkerThreadInitialization deleted 0x5ee3f8696a90  
UserRunAction deleted.  
UserPrimaryGenerator deleted.  
RunManager is deleting RunManagerKernel.  
G4SDManager deleted.  
EventManager deleted.  
Units table cleared.  
TransportationManager deleted.  
Total navigation history collections cleaned: 3  
G4RNGHelper object is deleted.  
===== Deleting memory pools =====  
Pool ID '20G4NavigationLevelRep', size : 0.00288 MB  
Pool ID '24G4ReferenceCountedHandleIvE', size : 0.000961 MB  
Pool ID '17G4DynamicParticle', size : 0.000961 MB  
Pool ID '15G4CountedObjectIvE', size : 0.000961 MB  
Number of memory pools allocated: 4; of which, static: 0  
Dynamic pools deleted: 4 / Total memory freed: 0.0058 MB  
===== G4Allocator objects are deleted.  
UIManager deleted.  
StateManager deleted.  
RunManagerKernel is deleted. Good bye :)  
RunManager is deleted.  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_b
```

Prova Esemplio B1

[illegible]

Ispezione log

```
sg@LAPTOP-4KU4SPH9: ~/Gt × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ ls  
CMakeCache.txt  Makefile          exampleB1      exampleB1.out  run1.log       run2.mac  
CMakeFiles      cmake_install.cmake exampleB1.in   init_vis.mac   run1.mac        vis.mac  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B1/B1_build$ more run1.log
```

Ispezione file log

```
===== Electromagnetic Physics Parameters =====
=====
LPM effect enabled 1
Enable creation and use of sampling tables 0
Apply cuts on all EM processes 0
Use combined TransportationWithMsc Disabled
Use general process 1
Enable linear polarisation for gamma 0
Enable photoeffect sampling below K-shell 1
Enable sampling of quantum entanglement 0
X-section factor for integral approach 0.8
Min kinetic energy for tables 100 eV
Max kinetic energy for tables 100 TeV
Number of bins per decade of a table 7
Verbose level 1
Verbose level for worker thread 0
Bremsstrahlung energy threshold above which
  primary e+- is added to the list of secondary 100 TeV
Bremsstrahlung energy threshold above which primary
  muon/hadron is added to the list of secondary 100 TeV
Lowest triplet kinetic energy 1 MeV
Enable sampling of gamma linear polarisation 0
5D gamma conversion model type 0
5D gamma conversion model on isolated ion 0
Livermore data directory epics_2017
```

Informazioni sui Parametri
Fisici dei diversi Processi

Ispezione file log

```
G4WT0 >
G4WT2 > * G4Track Information: Particle = gamma, Track ID = 1, Parent ID = 0
G4WT2 > *****
G4WT2 >
G4WT2 > Step#      X      Y      Z      KineE      dEStep      StepLeng      TrakLeng      Volume      Process
G4WT2 > 0    -1.42 cm -4.787 cm -15 cm      6 MeV      0 eV      0 fm      0 fm      Envelope      initStep
G4WT6 > Step#      X      Y      Z      KineE      dEStep      StepLeng      TrakLeng      Volume      Process
G4WT5 > ### Run 0 starts on worker thread 5.
G4WT2 > 1    -1.42 cm -4.787 cm -1.152 cm  1.643 MeV      0 eV     13.85 cm   13.85 cm   Envelope      compt
G4WT6 > 0    -5.483 cm -6.036 cm -13 cm    44.04 keV      0 eV      0 fm      0 fm      Envelope      initStep
G4WT2 > 2    -1.256 cm -5.174 cm -6.394 mm  286.9 keV      0 eV     6.627 mm   14.51 cm   Envelope      compt
G4WT2 > 3    -2.678 cm -2.784 cm -1.058 mm  233.6 keV      0 eV     2.832 cm   17.34 cm   Envelope      compt
G4WT6 > 1    -5.483 cm -6.038 cm -13 cm      0 eV     44.04 keV  34.84 um   34.84 um   Envelope      eIoni
G4WT6 >
G4WT0 > ### Run 0 starts on worker thread 0.
G4WT2 > 4    -6.633 cm -2.486 cm  1.489 cm  222.7 keV      0 eV     4.274 cm   21.62 cm   Envelope      compt
G4WT2 > 5     -10 cm -9.806 mm  4.926 cm  222.7 keV      0 eV     5.041 cm   26.66 cm   Envelope      Transportat
ion
G4WT2 > 6     -12 cm -862.4 um  6.967 cm  222.7 keV      0 eV     2.994 cm   29.65 cm   World      OutOfWorld
G4WT2 >
```

Info su Step
Traccia Primaria
(Track ID = 1,
Parent ID = 0)

```
G4WT2 > *****
G4WT2 > * G4Track Information: Particle = e-, Track ID = 7, Parent ID = 6
G4WT2 > *****
G4WT2 >
G4WT2 > Step#      X      Y      Z      KineE      dEStep      StepLeng      TrakLeng      Volume      Process
G4WT2 > 0    -1.563 cm -4.643 cm  1.709 mm  3.443 keV      0 eV      0 fm      0 fm      Envelope      initStep
G4WT2 > 1    -1.563 cm -4.643 cm  1.709 mm      0 eV     3.443 keV  409.5 nm   409.5 nm   Envelope      eIoni
G4WT6 > 0    -6.168 cm -4.924 cm -8.965 cm  4.975 MeV      0 eV      0 fm      0 fm      Envelope      initStep
G4WT6 > 1    -6.268 cm -4.951 cm -8.291 cm  3.466 MeV     1.509 MeV  7.175 mm   7.175 mm   Envelope      eIoni
G4WT6 > 2    -6.206 cm -5.057 cm -7.805 cm  2.433 MeV     1.033 MeV  5.363 mm   1.254 cm   Envelope      eIoni
G4WT6 > 3    -6.086 cm -5.153 cm -7.464 cm  1.771 MeV      662 keV   4.111 mm   1.665 cm   Envelope      eIoni
G4WT6 > 4    -5.986 cm -5.197 cm -7.194 cm  1.217 MeV     554.1 keV  3.298 mm   1.995 cm   Envelope      eIoni
G4WT6 > 5    -5.925 cm -5.194 cm -6.986 cm  833.3 keV     383.6 keV  2.599 mm   2.255 cm   Envelope      eIoni
G4WT6 > 6    -5.968 cm -5.227 cm -6.837 cm  405 keV      428.3 keV  2.079 mm   2.463 cm   Envelope      eIoni
G4WT6 > 7    -5.917 cm -5.276 cm -6.844 cm  117.1 keV     287.9 keV  1.258 mm   2.588 cm   Envelope      eIoni
G4WT6 > 8    -5.926 cm -5.278 cm -6.841 cm      0 eV     117.1 keV  188 um     2.607 cm   Envelope      eIoni
```

Info su Step
Altre Tracce
(Track ID > 1,
Parent ID > 0)

Esempio B4

Analisi Esempio B4

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.2.2/examples/basic
```

```
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$ pwd  
/home/sg/GEANT4  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$ ls  
geant4-v11.2.2 geant4-v11.2.2-build geant4-v11.2.2-install  
sg@LAPTOP-4KU4SPH9:~/GEANT4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4$ cd geant4-v11.2.2/examples/basic/  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$
```

Analisi Esempio B4

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.2.2/examples/basic  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$ ls  
B1 B2 B3 B4 B5 CMakeLists.txt GNUmakefile History README  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic$ more README
```


Analisi Esempio B4

ExampleB4

- Simplified calorimeter with layers of two materials
- Geometry with replica (G4PVReplica)
- Scoring within layers in four ways: via user actions (a),
via G4 sensitive detector and hits (c)
- Geant4 physics list (FTFP_BERT)
- Histograms (1D) and ntuple saved in the output file
- Started from novice/N03 example

Compilazione Esempio B4

```
sg@LAPTOP-4KU4SPH9: ~/Gt × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$ pwd  
/home/sg/GEANT4/geant4-v11.3.2/examples/basic/B4  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$ mkdir B4_build  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$
```

Compilazione Esempio B4

```
sg@LAPTOP-4KU4SPH9: ~/GI × + ▾
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$ cmake -DGeant4_DIR=/home/sg/GEANT4/geant4-v11.3.2
-install/lib/cmake/Geant4/ ../
CMake Warning (dev) in CMakeLists.txt:
  No project() command is present.  The top-level CMakeLists.txt file must
  contain a literal, direct call to the project() command.  Add a line of
  code such as

    project(ProjectName)

  near the top of the file, but after cmake_minimum_required().

  CMake is pretending there is a "project(Project)" command on the first
  line.
This warning is for project developers.  Use -Wno-dev to suppress it.

CMake Warning (dev) in CMakeLists.txt:
  cmake_minimum_required() should be called prior to this top-level project()
  call.  Please see the cmake-commands(7) manual for usage documentation of
  both commands.
This warning is for project developers.  Use -Wno-dev to suppress it.

-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
```

Compilazione Esempio B4a

```
sg@LAPTOP-4KU4SPH9: ~/Gi × + ▾
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$ ls
B4a  B4b  B4c  B4d  CMakeCache.txt  CMakeFiles  Makefile  cmake_install.cmake
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build$ cd B4a/
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$ ls
CMakeFiles  cmake_install.cmake  exampleB4a.out  init_vis.mac  plotNtuple.C  run2.mac
Makefile    exampleB4.in          gui.mac          plotHisto.C  run1.mac      vis.mac
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$ make -j 4 exampleB4a
[ 12%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/src/ActionInitialization.cc.o
[ 25%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/src/EventAction.cc.o
[ 37%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/src/DetectorConstruction.cc.o
[ 50%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/exampleB4a.cc.o
[ 62%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/src/PrimaryGeneratorAction.cc.o
[ 75%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/src/RunAction.cc.o
[ 87%] Building CXX object B4a/CMakeFiles/exampleB4a.dir/src/SteppingAction.cc.o
[100%] Linking CXX executable exampleB4a
[100%] Built target exampleB4a
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4_build/B4a$
```

Analisi Contenuto Esempio B4a

```
sg@LAPTOP-4KU4SPH9: ~/GI × + ▾  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$ ls  
B4_build B4a B4b B4c B4d CMakeLists.txt GNUmakefile History README macros  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4$ cd B4a/  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$ ls  
CMakeLists.txt exampleB4.in exampleB4a.out include plotHisto.C run1.mac src  
GNUmakefile exampleB4a.cc gui.mac init_vis.mac plotNtuple.C run2.mac vis.mac  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$ ls -l include/  
ActionInitialization.hh  
DetectorConstruction.hh  
EventAction.hh  
PrimaryGeneratorAction.hh  
RunAction.hh  
SteppingAction.hh  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$ ls -l src/  
ActionInitialization.cc  
DetectorConstruction.cc  
EventAction.cc  
PrimaryGeneratorAction.cc  
RunAction.cc  
SteppingAction.cc  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.3.2/examples/basic/B4/B4a$
```

Esempio B4a – main()

exampleB4a.cc

```
// Construct the default run manager
//
auto runManager =
    G4RunManagerFactory::CreateRunManager(G4RunManagerType::Default);
#ifdef G4MULTITHREADED
    if ( nThreads > 0 ) {
        runManager->SetNumberOfThreads(nThreads);
    }
#endif

// Set mandatory initialization classes
//
auto detConstruction = new B4::DetectorConstruction();
runManager->SetUserInitialization(detConstruction);

auto physicsList = new FTFP_BERT;
runManager->SetUserInitialization(physicsList);

auto actionInitialization = new B4a::ActionInitialization(detConstruction);
runManager->SetUserInitialization(actionInitialization);

if ( macro.size() ) {
    // batch mode
    G4String command = "/control/execute ";
    UImanager->ApplyCommand(command+macro);
}
else {
    // interactive mode : define UI session
    UImanager->ApplyCommand("/control/execute init_vis.mac");
    if (ui->IsGUI()) {
        UImanager->ApplyCommand("/control/execute gui.mac");
    }
    ui->SessionStart();
    delete ui;
}
```


Detector Construction B4a

DetectorConstruction.hh

```
/// Detector construction class to define materials and geometry.
/// The calorimeter is a box made of a given number of layers. A layer consists
/// of an absorber plate and of a detection gap. The layer is replicated.
///
/// Four parameters define the geometry of the calorimeter :
///
/// - the thickness of an absorber plate,
/// - the thickness of a gap,
/// - the number of layers,
/// - the transverse size of the calorimeter (the input face is a square).
///
/// In addition a transverse uniform magnetic field is defined
/// via G4GlobalMagFieldMessenger class.

class DetectorConstruction : public G4VUserDetectorConstruction
```

DetectorConstruction.cc

```
G4VPhysicalVolume* DetectorConstruction::Construct()
{
    // Define materials
    DefineMaterials();

    // Define volumes
    return DefineVolumes();
}
```

Detector Construction B4a

```
void DetectorConstruction::DefineMaterials()
{
    // Lead material defined using NIST Manager
    auto nistManager = G4NistManager::Instance();
    nistManager->FindOrBuildMaterial("G4_Pb");

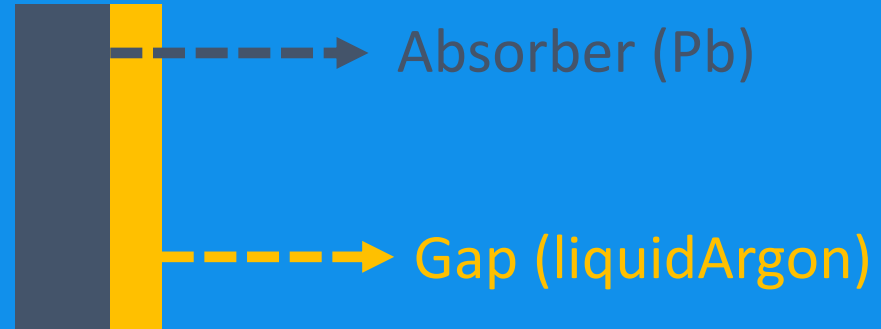
    // Liquid argon material
    G4double a; // mass of a mole;
    G4double z; // z=mean number of protons;
    G4double density;
    new G4Material("liquidArgon", z=18., a= 39.95*g/mole, density= 1.390*g/cm3);
    // The argon by NIST Manager is a gas with a different density

    // Vacuum
    new G4Material("Galactic", z=1., a=1.01*g/mole,density= universe_mean_density,
        kStateGas, 2.73*kelvin, 3.e-18*pascal);

    // Print materials
    G4cout << *(G4Material::GetMaterialTable()) << G4endl;
}
```

```
G4VPhysicalVolume* DetectorConstruction::DefineVolumes()
```

Layer



Calorimeter

(Serie di Repliche del volume Layer)



ActionInitialization / PrimaryGeneratorAction

```
void ActionInitialization::Build() const
{
    SetUserAction(new PrimaryGeneratorAction);
    SetUserAction(new RunAction);
    auto eventAction = new EventAction;
    SetUserAction(eventAction);
    SetUserAction(new SteppingAction(fDetConstruction, eventAction));
}
```

ActionInitialization.cc

PrimaryGeneratorAction.hh/cc

Particelle Primarie

Informazioni su Interazioni Particelle

Per capire come le informazioni sull'interazione fra particelle e rivelatore vengano recuperate e immagazzinate bisogna analizzare:

- RunAction.cc

AnalysisManager (salvataggio info su ROOT)

- EventAction.cc

Inizializzazione Salvataggio Edep/Evento

- SteppingAction.cc

Recupero Info su Interazioni Particlle / Step

RunAction

Costruttore: Analysis Manager

```
RunAction::RunAction()
{
    // set printing event number per each event
    G4RunManager::GetRunManager()->SetPrintProgress(1);

    // Create analysis manager
    // The choice of the output format is done via the specified
    // file extension.
    auto analysisManager = G4AnalysisManager::Instance();
```

Costruttore: Istogrammi e Ntuple ROOT

```
// Creating histograms
analysisManager->CreateH1("Eabs" , "Edep in absorber", 110, 0., 330*MeV);
analysisManager->CreateH1("Egap" , "Edep in gap", 100, 0., 30*MeV);
analysisManager->CreateH1("Labs" , "trackL in absorber", 100, 0., 50*cm);
analysisManager->CreateH1("Lgap" , "trackL in gap", 100, 0., 50*cm);

// Creating ntuple
//
analysisManager->CreateNtuple("B4", "Edep and TrackL");
analysisManager->CreateNtupleDColumn("Eabs");
analysisManager->CreateNtupleDColumn("Egap");
analysisManager->CreateNtupleDColumn("Labs");
analysisManager->CreateNtupleDColumn("Lgap");
analysisManager->FinishNtuple();
```

BeginOfRunAction: Apertura/Creazione Output File

```
void RunAction::BeginOfRunAction(const G4Run* /*run*/)
{
    //inform the runManager to save random number seed
    //G4RunManager::GetRunManager()->SetRandomNumberStore(true);

    // Get analysis manager
    auto analysisManager = G4AnalysisManager::Instance();

    // Open an output file
    //
    G4String fileName = "B4.root";
    // Other supported output types:
    // G4String fileName = "B4.csv";
    // G4String fileName = "B4.hdf5";
    // G4String fileName = "B4.xml";
    analysisManager->OpenFile(fileName);
```

EndOfRunAction: Salvataggio Dati / Chiusura Output File

```
void RunAction::EndOfRunAction(const G4Run* /*run*/)
{
    // save histograms & ntuple
    //
    analysisManager->Write();
    analysisManager->CloseFile();
}
```

EventAction

EventAction.cc / BeginOfEventAction:
Inizializzazione Variabili per Immagazzinare Info Particelle

```
void EventAction::BeginOfEventAction(const G4Event* /*event*/)
{
    // initialisation per event
    fEnergyAbs = 0.;
    fEnergyGap = 0.;
    fTrackLAbs = 0.;
    fTrackLGap = 0.;
}
```

EventAction.cc / EndOfEventAction:
Riempimento Istogrammi e Contenuto Ntupla

```
void EventAction::EndOfEventAction(const G4Event* event)
{
    // get analysis manager
    auto analysisManager = G4AnalysisManager::Instance();

    // fill histograms
    analysisManager->FillH1(0, fEnergyAbs);
    analysisManager->FillH1(1, fEnergyGap);
    analysisManager->FillH1(2, fTrackLAbs);
    analysisManager->FillH1(3, fTrackLGap);

    // fill ntuple
    analysisManager->FillNtupleDColumn(0, fEnergyAbs);
    analysisManager->FillNtupleDColumn(1, fEnergyGap);
    analysisManager->FillNtupleDColumn(2, fTrackLAbs);
    analysisManager->FillNtupleDColumn(3, fTrackLGap);
    analysisManager->AddNtupleRow();
}
```

EventAction.hh :
Accumulo Info Particelle Step

```
// inline functions

inline void EventAction::AddAbs(G4double de, G4double dl) {
    fEnergyAbs += de;
    fTrackLAbs += dl;
}

inline void EventAction::AddGap(G4double de, G4double dl) {
    fEnergyGap += de;
    fTrackLGap += dl;
}
```


SteppingAction

UserSteppingAction

```
void SteppingAction::UserSteppingAction(const G4Step* step)
{
    // Collect energy and track length step by step

    // get volume of the current step
    auto volume = step->GetPreStepPoint()->GetTouchableHandle()->GetVolume();

    // energy deposit
    auto edep = step->GetTotalEnergyDeposit();

    // step length
    G4double stepLength = 0.;
    if ( step->GetTrack()->GetDefinition()->GetPDGCharge() != 0. ) {
        stepLength = step->GetStepLength();
    }

    if ( volume == fDetConstruction->GetAbsorberPV() ) {
        fEventAction->AddAbs(edep,stepLength);
    }

    if ( volume == fDetConstruction->GetGapPV() ) {
        fEventAction->AddGap(edep,stepLength);
    }
}
```

Energia Depositata
nello Step

Volume all'interno di cui
avviene il processo

Edep, StepLength
per Assorbitore Pb

Lunghezza Step (solo per
particelle cariche)

Edep, StepLength per Gap
Argon Liquido

Informazioni su Interazioni Particelle

RunAction

Creazione File Output e Ntupla ROOT

Ciclo Eventi

EventAction

Inizializzazione Informazioni su Edep / Lunghezza Traccia

Ciclo Step

SteppingAction

Recupero Info su Edep / Lunghezza Traccia per ogni Step

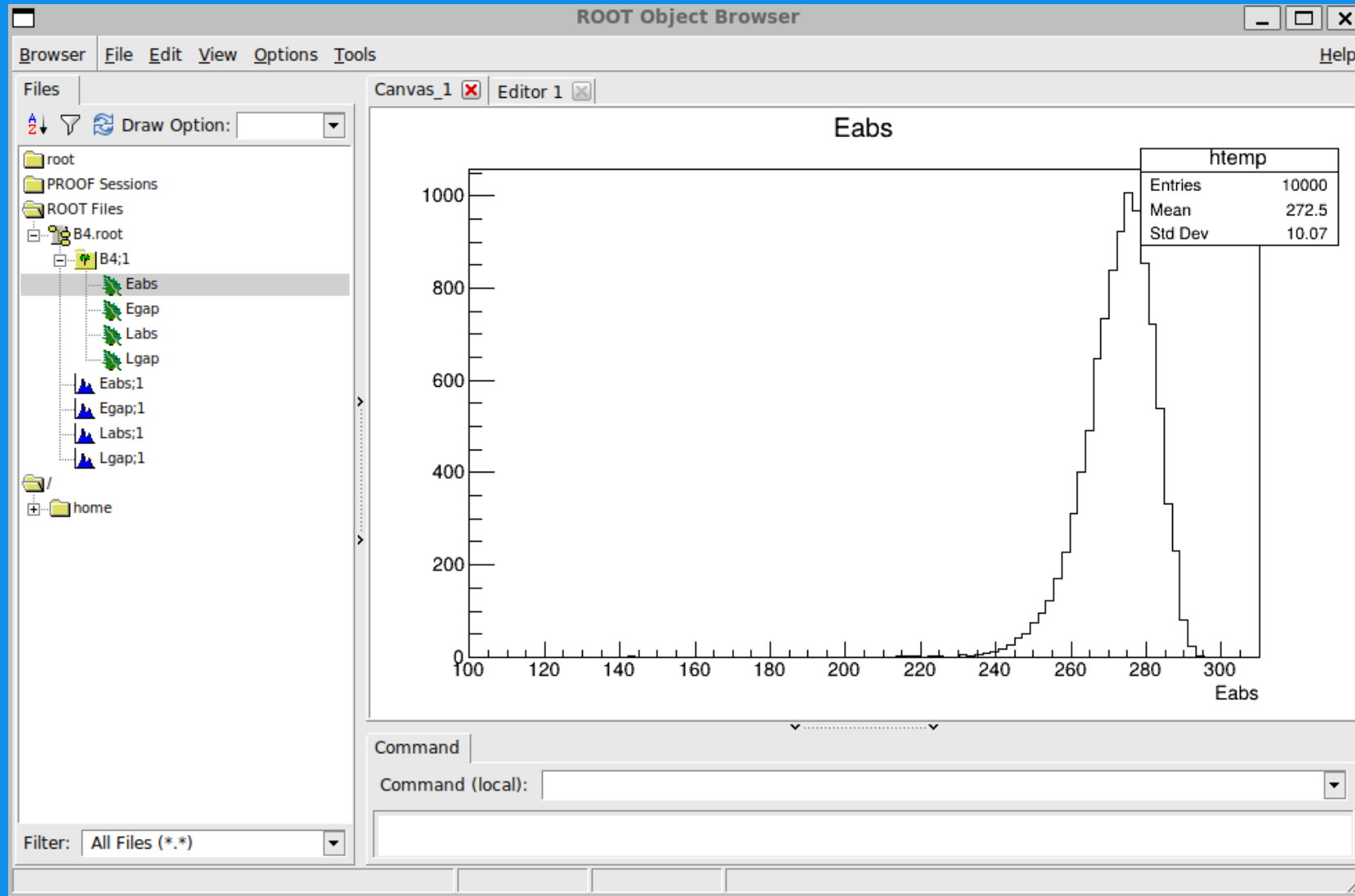
EventAction

Accumulo Informazione su Edep / Lunghezza Traccia

RunAction

Salvataggio File Output

ROOT Output File



Analisi Contenuto Esempio B4c

```
sg@LAPTOP-4KU4SPH9: ~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$ ls -1 include/
ActionInitialization.hh
CalorHit.hh
CalorimeterSD.hh
DetectorConstruction.hh
EventAction.hh
PrimaryGeneratorAction.hh
RunAction.hh
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$ ls -1 src/
ActionInitialization.cc
CalorHit.cc
CalorimeterSD.cc
DetectorConstruction.cc
EventAction.cc
PrimaryGeneratorAction.cc
RunAction.cc
sg@LAPTOP-4KU4SPH9:~/GEANT4/geant4-v11.2.2/examples/basic/B4/B4c$
```

NO
SteppingAction

CalorHit

CalorHit.hh

```
/// Calorimeter hit class
///
/// It defines data members to store the the energy deposit and track lengths
/// of charged particles in a selected volume:
/// - fEdep, fTrackLength

class CalorHit : public G4VHit
{
public:
    CalorHit() = default;
    CalorHit(const CalorHit&) = default;
    ~CalorHit() override = default;

    // methods to handle data
    void Add(G4double de, G4double dl);

    // get methods
    G4double GetEdep() const;
    G4double GetTrackLength() const;

private:
    G4double fEdep = 0.;      ///< Energy deposit in the sensitive volume
    G4double fTrackLength = 0.; ///< Track length in the sensitive volume
};

//....ooo000ooo.....ooo000ooo.....ooo000ooo.....ooo000ooo.....

using CalorHitsCollection = G4THitsCollection<CalorHit>;

extern G4ThreadLocal G4Allocator<CalorHit>* CalorHitAllocator;
```

Sensitive Detector

CalorimeterSD.hh

```
/// Calorimeter sensitive detector class
///
/// In Initialize(), it creates one hit for each calorimeter layer and one more
/// hit for accounting the total quantities in all layers.
///
/// The values are accounted in hits in ProcessHits() function which is called
/// by Geant4 kernel at each step.

class CalorimeterSD : public G4VSensitiveDetector
{
public:
    CalorimeterSD(const G4String& name,
                  const G4String& hitsCollectionName,
                  G4int nofCells);
    ~CalorimeterSD() override = default;

    // methods from base class
    void Initialize(G4HCOFThisEvent* hitCollection) override;
    G4bool ProcessHits(G4Step* step, G4TouchableHistory* history) override;
    void EndOfEvent(G4HCOFThisEvent* hitCollection) override;

private:
    CalorHitsCollection* fHitsCollection = nullptr;
    G4int fNofCells = 0;
};
```

Sensitive Detector

CalorimeterSD.cc

```
CalorimeterSD::CalorimeterSD(const G4String& name,
                             const G4String& hitsCollectionName,
                             G4int nofCells)
: G4VSensitiveDetector(name),
  fNofCells(nofCells)
{
  collectionName.insert(hitsCollectionName);
}

void CalorimeterSD::Initialize(G4HCofThisEvent* hce)
{
  // Create hits collection
  fHitsCollection
    = new CalorHitsCollection(SensitiveDetectorName, collectionName[0]);

  // Add this collection in hce
  auto hcID
    = G4SDManager::GetSDMpointer()->GetCollectionID(collectionName[0]);
  hce->AddHitsCollection( hcID, fHitsCollection );

  // Create hits
  // fNofCells for cells + one more for total sums
  for (G4int i=0; i<fNofCells+1; i++ ) {
    fHitsCollection->insert(new CalorHit());
  }
}
```


Sensitive Detector

CalorimeterSD.cc

```
G4bool CalorimeterSD::ProcessHits(G4Step* step,
                                   G4TouchableHistory*)
{
    // energy deposit
    auto edep = step->GetTotalEnergyDeposit();

    // step length
    G4double stepLength = 0.;
    if ( step->GetTrack()->GetDefinition()->GetPDGCharge() != 0. ) {
        stepLength = step->GetStepLength();
    }

    if ( edep==0. && stepLength == 0. ) return false;

    auto touchable = (step->GetPreStepPoint()->GetTouchable());

    // Get calorimeter cell id
    auto layerNumber = touchable->GetReplicaNumber(1);

    // Get hit accounting data for this cell
    auto hit = (*fHitsCollection)[layerNumber];
    if ( ! hit ) {
        G4ExceptionDescription msg;
        msg << "Cannot access hit " << layerNumber;
        G4Exception("CalorimeterSD::ProcessHits()",
                    "MyCode0004", FatalException, msg);
    }

    // Get hit for total accounting
    auto hitTotal
        = (*fHitsCollection)[fHitsCollection->entries()-1];

    // Add values
    hit->Add(edep, stepLength);
    hitTotal->Add(edep, stepLength);
}
```

Detector Construction + Sensitive Detector

```
void DetectorConstruction::ConstructSDandField()
{
    // G4SDManager::GetSDMpointer()->SetVerboseLevel(1);

    //
    // Sensitive detectors
    //
    auto absoSD
        = new CalorimeterSD("AbsorberSD", "AbsorberHitsCollection", fNofLayers);
    G4SDManager::GetSDMpointer()->AddNewDetector(absoSD);
    SetSensitiveDetector("AbsorberSD", absoSD);

    auto gapSD
        = new CalorimeterSD("GapSD", "GapHitsCollection", fNofLayers);
    G4SDManager::GetSDMpointer()->AddNewDetector(gapSD);
    SetSensitiveDetector("GapSD", gapSD);

    //
    // Magnetic field
    //
    // Create global magnetic field messenger.
    // Uniform magnetic field is then created automatically if
    // the field value is not zero.
    G4ThreeVector fieldValue;
    fMagFieldMessenger = new G4GlobalMagFieldMessenger(fieldValue);
    fMagFieldMessenger->SetVerboseLevel(1);
}
```

EventAction

```
CalorHitsCollection*
EventAction::GetHitsCollection(G4int hcID,
                               const G4Event* event) const
{
    auto hitsCollection
    = static_cast<CalorHitsCollection*>(
        event->GetHCofThisEvent()->GetHC(hcID));

    if ( ! hitsCollection ) {
        G4ExceptionDescription msg;
        msg << "Cannot access hitsCollection ID " << hcID;
        G4Exception("EventAction::GetHitsCollection()",
                    "MyCode0003", FatalException, msg);
    }

    return hitsCollection;
}
```

EventAction

```
void EventAction::EndOfEventAction(const G4Event* event)
{
    // Get hits collections IDs (only once)
    if ( fAbsHCID == -1 ) {
        fAbsHCID
            = G4SDManager::GetSDMpointer()->GetCollectionID("AbsorberHitsCollection");
        fGapHCID
            = G4SDManager::GetSDMpointer()->GetCollectionID("GapHitsCollection");
    }

    // Get hits collections
    auto absoHC = GetHitsCollection(fAbsHCID, event);
    auto gapHC = GetHitsCollection(fGapHCID, event);

    // Get hit with total values
    auto absoHit = (*absoHC)[absoHC->entries()-1];
    auto gapHit = (*gapHC)[gapHC->entries()-1];

    // fill ntuple
    analysisManager->FillNtupleDColumn(0, absoHit->GetEdep());
    analysisManager->FillNtupleDColumn(1, gapHit->GetEdep());
    analysisManager->FillNtupleDColumn(2, absoHit->GetTrackLength());
    analysisManager->FillNtupleDColumn(3, gapHit->GetTrackLength());
    analysisManager->AddNtupleRow();
}
```

Esercizio

Muoni in Acqua

Partendo dagli esempi esaminati sviluppare un progetto come segue:

- Definire un rivelatore consistente in un cilindro di acqua con le seguenti dimensioni:
 - Raggio 5 m
 - Altezza 5 m
- Salvare in un file ROOT le seguenti informazioni:
 - Accumulo energia depositata e lunghezza percorso nel rivelatore per la traccia primaria
 - Accumulo energia depositata e lunghezza percorso nel rivelatore per ogni traccia nell'evento
- Generare eventi primari con muoni da 1 GeV
- Analizzare Energia Depositata vs Lunghezza Traccia Primaria

Suggerimenti

- Creare una cartella apposita per l'esercizio (Es. /home/user/GEANT4/Esercizio)
- Copiare GNUmakefile e CMakeList.txt da Esempio B1
- Copiare file main e cartelle include e src da Esempio B4
- Copiare macro per esecuzione da Esempio B4
- Modificare nomi e contenuto file appropriatamente

```
sg@LAPTOP-4KU4SPH9: ~/GI x + v
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$ ls
CMakeLists.txt GNUmakefile build esercizioG4.cc include src
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$ ls -1 include/
ActionInitialization.hh
DetectorConstruction.hh
EventAction.hh
PrimaryGeneratorAction.hh
RunAction.hh
SteppingAction.hh
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$ ls -1 src/
ActionInitialization.cc
DetectorConstruction.cc
EventAction.cc
PrimaryGeneratorAction.cc
RunAction.cc
SteppingAction.cc
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
```



```
sg@LAPTOP-4KU4SPH9: ~/GI × + ▾
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi$ cd build/
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ cmake -DGeant4_DIR=/home/sg/GEANT4/geant4-v11.3.2-install/lib/cmake/Geant4/
../
-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Found EXPAT: /usr/lib/x86_64-linux-gnu/libexpat.so (found suitable version "2.6.1", minimum required is "2.6.1")
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- Found Geant4: /home/sg/GEANT4/geant4-v11.3.2-install/lib/cmake/Geant4/Geant4Config.cmake (found version "11.3.2")
-- Configuring done (1.6s)
-- Generating done (0.0s)
-- Build files have been written to: /home/sg/GEANT4/Esercizi/build
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
```



```
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ ls -l  
CMakeCache.txt  
CMakeFiles  
Makefile  
cmake_install.cmake  
esercizioG4  
run10.mac  
run10000.mac  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
```

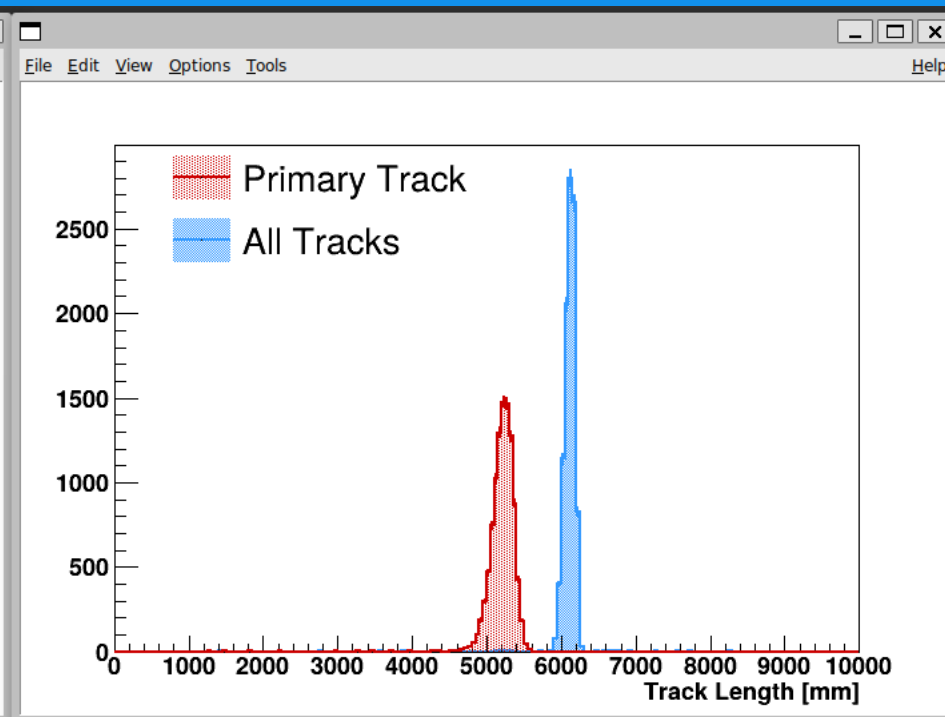
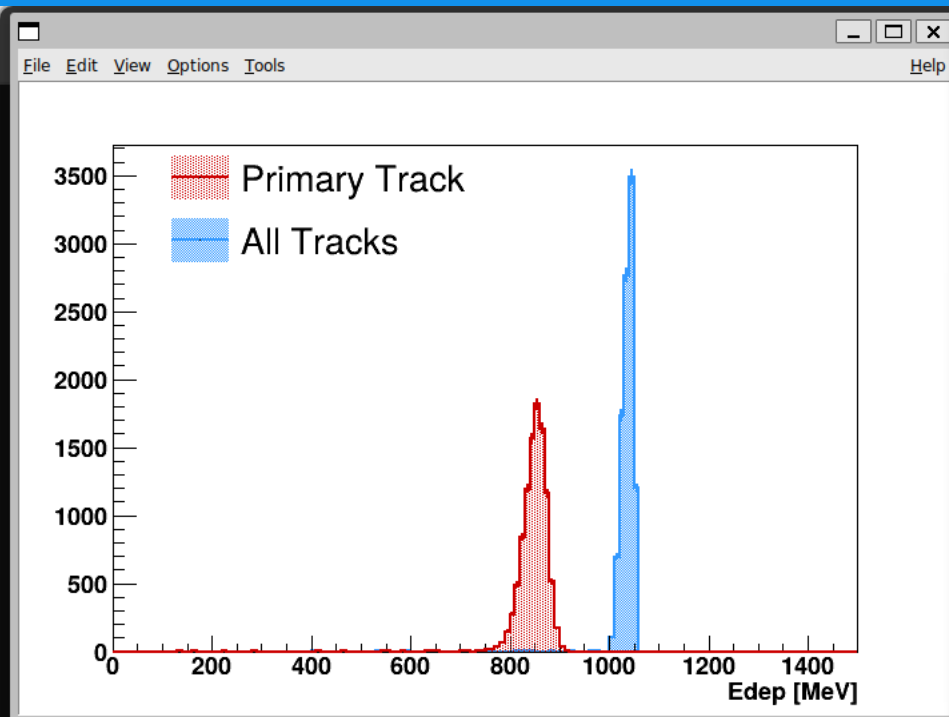
```
sg@LAPTOP-4KU4SPH9: ~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ more run10000.mac  
# Macro file for EsercizioG4  
#  
# Can be run in batch, without graphic  
# or interactively: Idle> /control/execute run1.mac  
#  
# change the default number of workers (in multi-threading mode)  
#/run/numberOfThreads 4  
#  
# Kind of tutorial:  
# interactively with visualization, issue the commands one by one:  
# Idle> gun/particle mu+  
# Idle> ...etc...  
#  
# Initialize kernel  
/run/initialize  
#  
# muon 1000 MeV in direction (0.,0.,1.)  
/gun/particle mu+  
/gun/energy 1000 MeV  
#  
/run/printProgress 5  
/run/beamOn 10000  
#  
# verbosity  
#/tracking/verbose 2  
#/run/beamOn 1  
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ |
```

```
sg@LAPTOP-4KU4SPH9: ~/GI × + ▾
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ ./esercizioG4 -m run10000.mac > run10000.log
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ ls -l
CMakeCache.txt
CMakeFiles
EsercizioG4.root
Makefile
cmake_install.cmake
esercizioG4
run10.mac
run10000.log
run10000.mac
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
```

```

sg@LAPTOP-4KU4SPH9: ~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/build$ root -l EsercizioG4.root
root [0]
Attaching file EsercizioG4.root as _file0...
(TFile *) 0x63e070c9d000
root [1] .ls
TFile**      EsercizioG4.root
TFile*       EsercizioG4.root
  KEY: TTree  EsercizioG4;1  Edep and TrackL
  KEY: TH1D   Eprm;1  Edep for Primary Track
  KEY: TH1D   Eall;1  Edep for All Tracks
  KEY: TH1D   Lprm;1  trackL for Primatry Track
  KEY: TH1D   Lall;1  trackL for All Tracks
root [2] |

```



```
sg@LAPTOP-4KU4SPH9:~/GEANT4/Esercizi/root$ root EsercizioG4.C
```

```
-----  
| Welcome to ROOT 6.36.04                               https://root.cern |  
| (c) 1995-2025, The ROOT Team; conception: R. Brun, F. Rademakers |  
| Built for linuxx8664gcc on Aug 25 2025, 09:43:16 |  
| From tags/v6-36-04@v6-36-04 |  
| With c++ (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0 |  
| Try '.help'/'?', '.demo', '.license', '.credits', '.quit'/'q' |  
-----
```

```
root [0]  
Processing EsercizioG4.C...  
root [1]
```

Repository GitHub

<https://github.com/s-germani/geant4-tecniche-sperimentali-2025>

geant4-tecniche-sperimentali-2025

Documentazione per le lezioni su Geant4 del corso di "Tecniche Sperimentali per la Fisica delle Alte Eenergie" per la Laurea Magistrale in Fisica dell'Università degli Studi di Perugia.

- le lezioni sono disponibili sotto forma di file PDF;
- è disponibile anche il codice per la soluzine dell'esercizio proposto.

Utilizzo soluzioni:

1. Nella cartella `Esercizio` creare la sottocartella `build` e spostarsi al suo interno:

```
$ mkdir build $ cd build
```

2. compilare il codice attraverso i comandi:

```
$ cmake -DGeant4_DIR=/path/to/geant4-install/lib/cmake/Geant4/ .. $ make -j 4 esercizioG4
```

3. eseguire con:

```
$ ./esercizioG4 -m run10000.mac > run10000.log
```

Un esempio di macro ROOT per analizzare il file generato è disponibile nella cartella `root`.